

E-COMMERCE TRANSACTIONS DATASET

Cleaning Challenges

- CustomerID had 135,080 missing values — filled with 0 to avoid losing transaction data
- InvoiceDate was stored as a string — converted to datetime
- 10,587 rows had negative quantity values indicating returns
- 9,251 cancelled invoices identified by 'C' prefix in InvoiceNo
- 5,268 duplicate rows were found and removed

Key EDA Findings

- Data covers December 2010 to December 2011
- United Kingdom dominates revenue at £9 million
- Revenue peaks in November 2011 due to holiday shopping season
- Most products are priced below £5

Top Insights

1. United Kingdom is the primary market generating over 90% of revenue
2. Revenue surges in Q4 confirming strong holiday shopping behavior
3. A small group of wholesale customers drive a large portion of revenue